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Tippecanoe High School

Science Fair Final Report

Ion the Prize: Turbojet Efficiency

One of the critical challenges facing our society today is the transition away from fossil fuels. My project aims to facilitate this transition by exploring the effects of feeding ionized air into a model turbojet engine. This study will investigate how this process impacts the engine. What are the effects of feeding ionized air into a model turbojet engine? How does feeding ionized air into a turbojet engine impact its power and efficiency? What changes occur in the internal temperature of a turbojet engine when ionized air is fed into it? What adverse effects, such as accelerated corrosion, might arise in a turbojet engine after feeding it ionized air? These are just a few questions that I hope to answer with this project.

In order to understand this project, it is helpful to know how a turbojet engine works. In a turbojet, the intake air is first compressed, then mixed with fuel in the combustion chamber, then used to drive the turbine. The driving force on this turbine is in turn used to power the compressor, sustaining the process. Since commercial air travel is almost always powered by a turbine engine (turbofan, similar to a turbojet), about 2.5% of carbon emissions annually are attributed to aviation (Ritchie). Understanding that a complete transition to a more sustainable power plant for aviation is not feasible at this time, I came to the conclusion that improving the efficiency of a turbine engine could be a solution. Lockheed Martin has patented (US5692481A) a process of ionizing air via high-energy light to accomplish this (Miller); however, I plan to take a different approach. By utilizing high-voltage ion thrusters to ionize intake air, combustion can be enhanced; this has been tested and patented many times for use on internal combustion piston

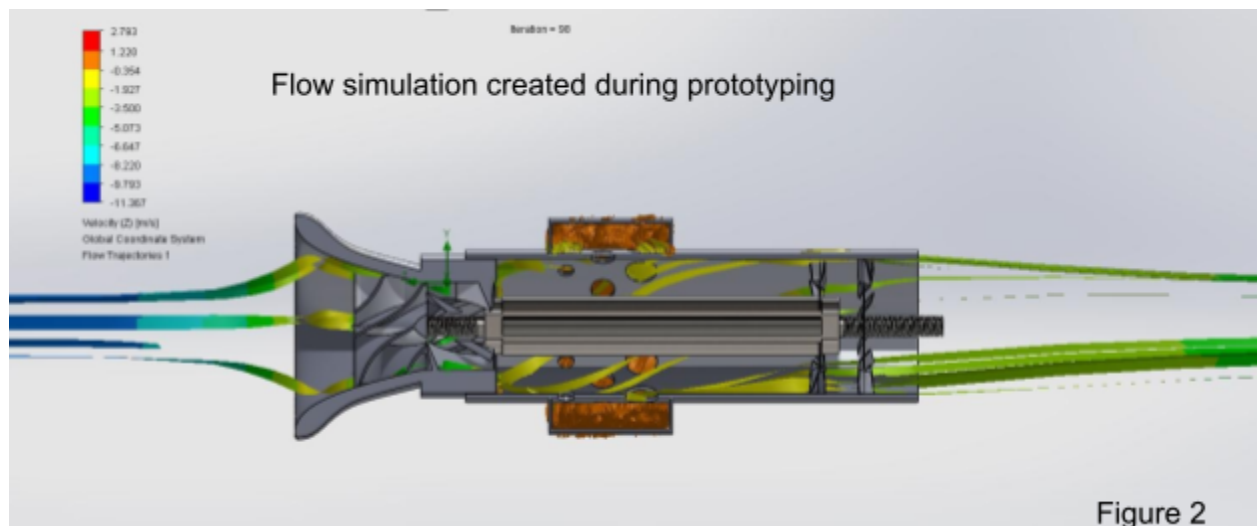
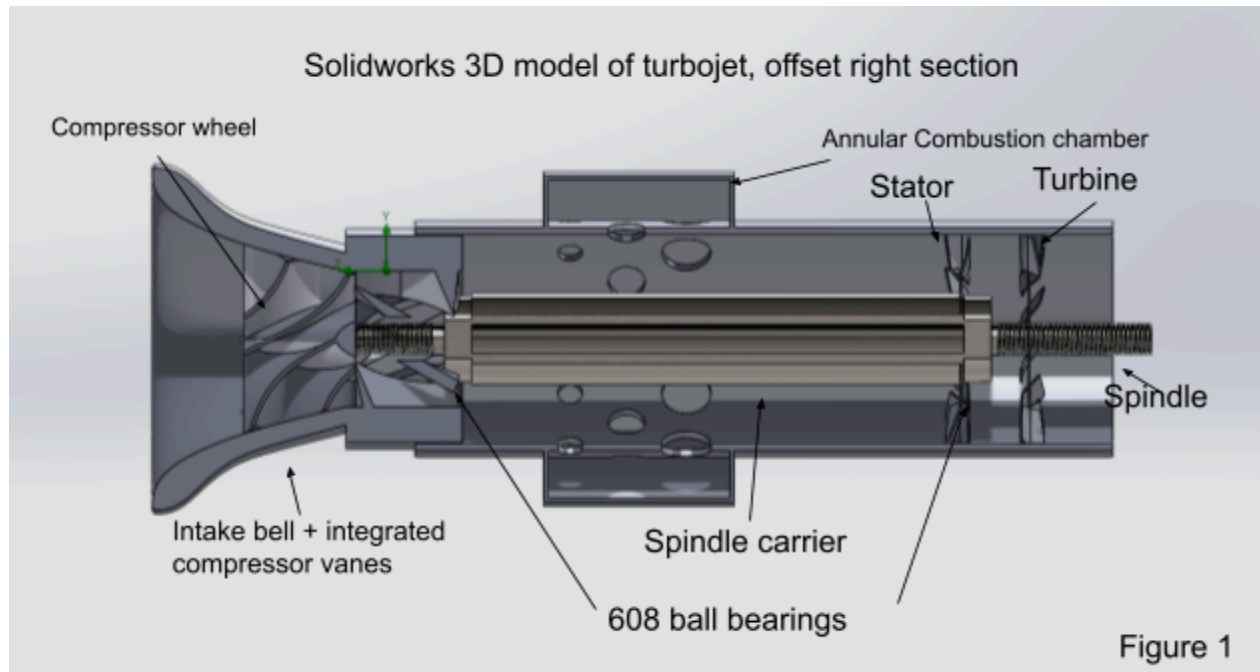
engines. One specific example is US5010869A. More complete combustion can be achieved since the ionization of the air results in ozone, which is much more reactive than diatomic oxygen. The intent is to reduce harmful emissions such as nitrogen oxides, carbon monoxide, and others. This project intends to analyze the extent of emission reductions and all other effects this process may have, positive or negative.

To analyze these effects, I first needed to create a model turbojet engine. To accomplish this, I utilized 2 ¼ ID steel tubing for a housing, a 3.4-inch diameter steel can for a combustion chamber, Inconel 718 alloy washers for single stage turbine stator and fan combo, 220mm threaded rod for the spindle, modified 608 bearings, 1 inch EMT conduit for the spindle housing, and designed/3d printed the compressor, intake bell, and intake vanes all in a high temp PETG filament. Material choices varied depending on the stage of the engine, inconel was chosen for the turbine due to its high heat tolerance whilst PETG components were made for the intake due to ease of manufacturing. For this step of my project, I used propane as a fuel; however, I have made arrangements to hopefully test with kerosene in the future. The ion engine is also self-designed and printed with off-the-shelf components and aluminum sheet. I utilized 400kV-rated DC voltage multipliers as a power source to create the plasma needed. In testing, I conducted a series of six, 30-second trials. Each test consisted of six trials, Test 1 with normal atmosphere, and Test 2 with ionized air. All tests were conducted at 5°C or 6°C ambient temperature and within four 4 hours of each other. Data points that were collected are internal temperature(°C), the thrust produced (g), oxidizing/combustible exhaust gases (qualitative in V), and power used by the ion thruster when applicable. Sensors+breakouts included: 500°C K-type thermocouple+MAX31855, MiCS5524 gas sensor, 1Kg load cell+HX711 ADC, and FCS521-SD-5V respectively. In addition to measuring these parameters, I tore down and inspected the turbine engine for adverse reactions to the ionized gases.

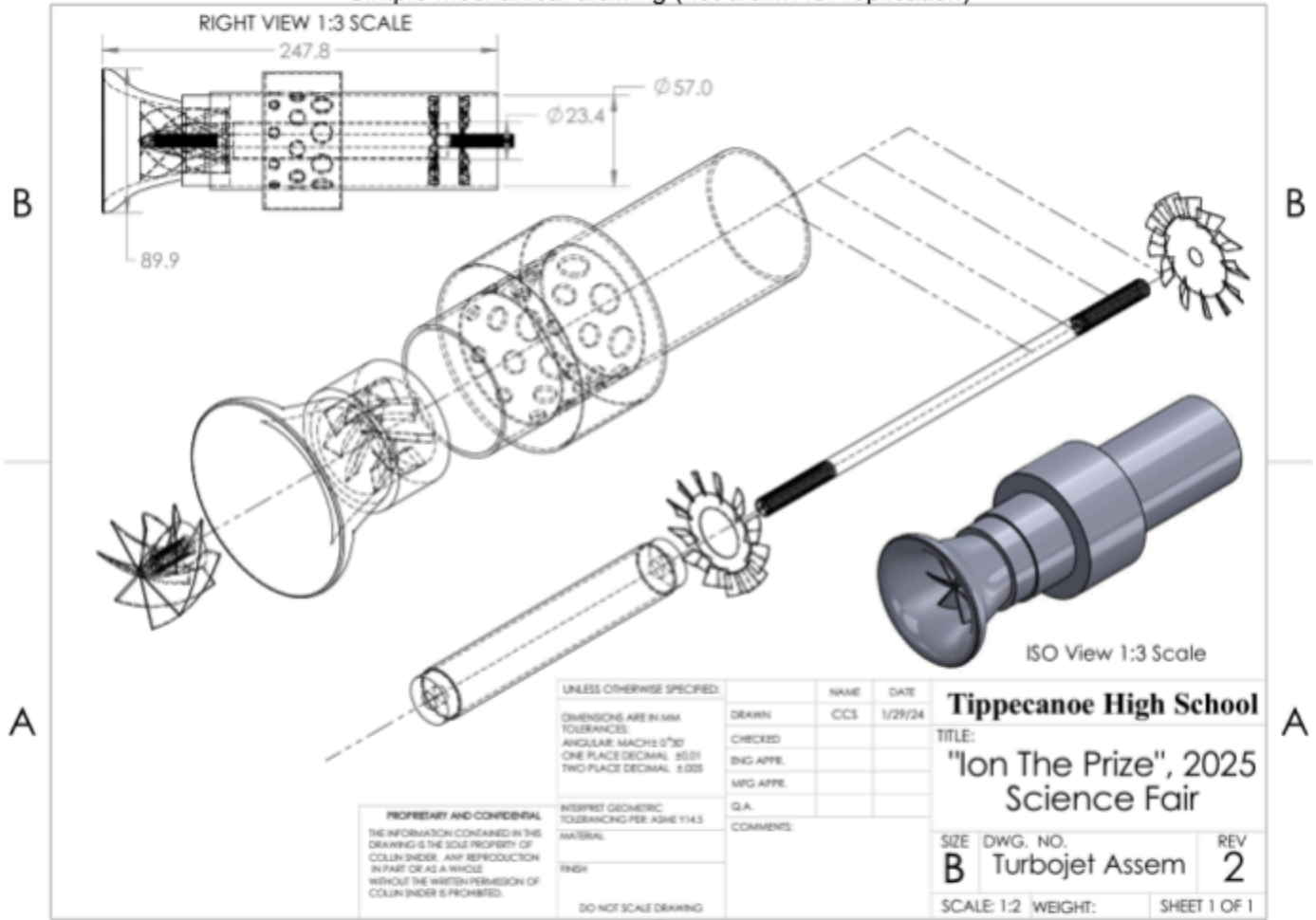
To conduct the trials, the engine was first “spooled up” to starting rpm, the ion thruster turned on(when applicable), the gas turned on, and the igniter engaged for a moment, then reduce cooling air flow to 0.5 psi. Subsequently, the serial monitor on the laptop was reset to record new data for 30 seconds. After data was collected, the engine was opened, re-lubricated, and allowed to cool to below 30°C. This was performed six times in total. It is acknowledged that fire and non-lethal electric shock were possible in testing and proper precautions and cautions such as a fire plan and PPE were taken.

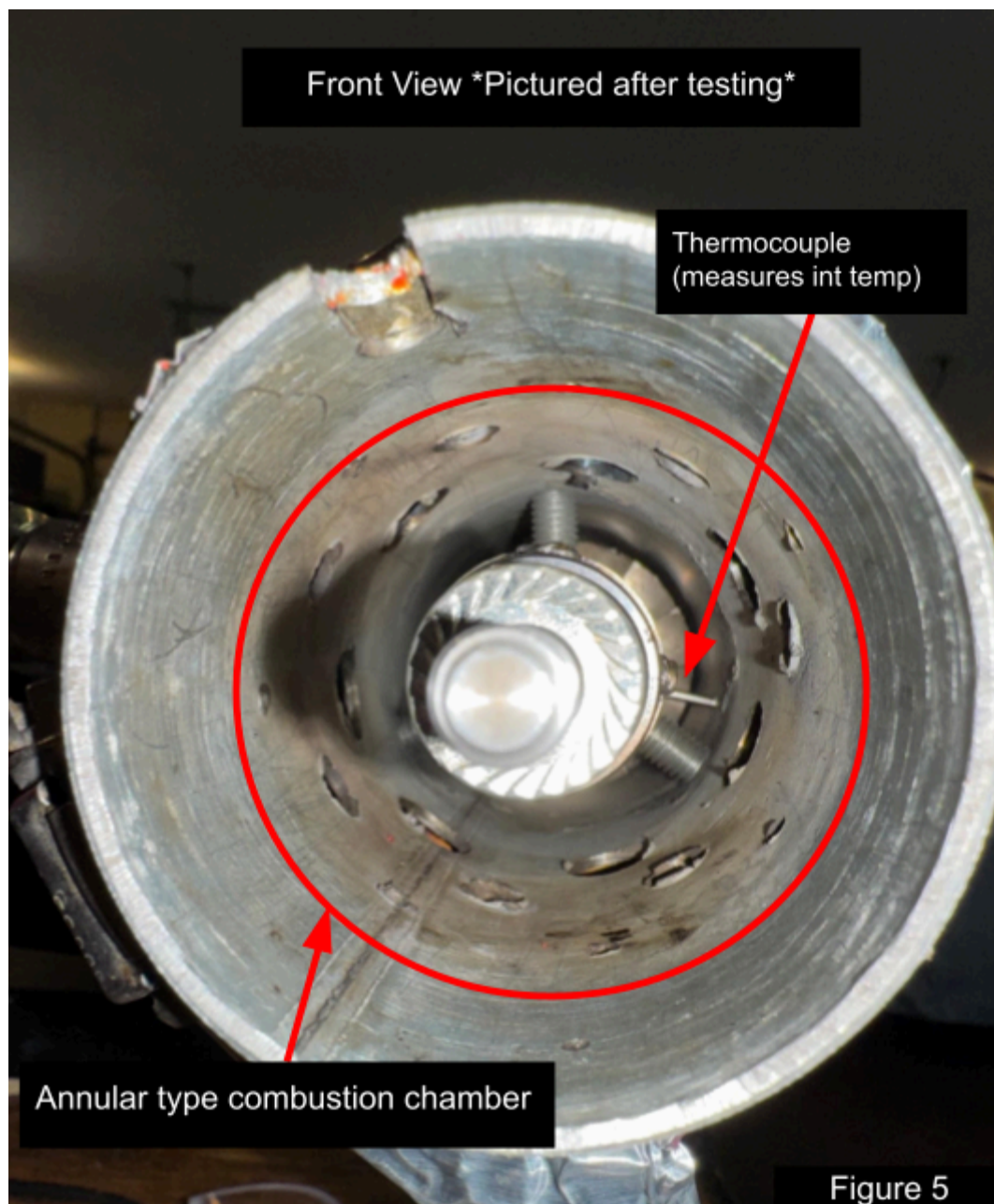
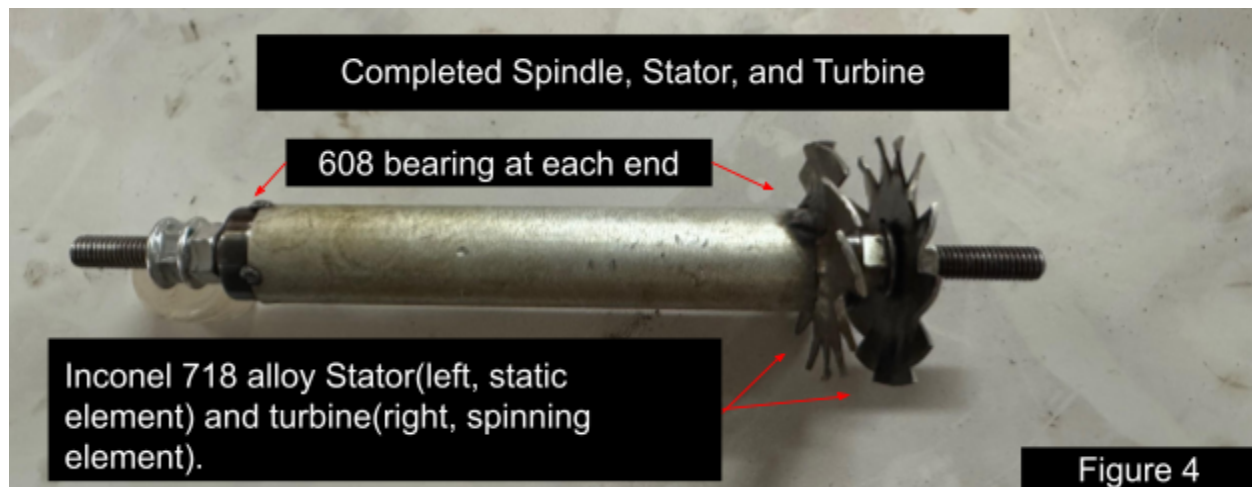
With prior knowledge and research into the topic such as reading patents, papers, and other tests, I hypothesize that the engine being fed ionized air will have a reduction in harmful exhaust emissions, an increase in combustion temperature, and an increase in thrust, in comparison to the control. I predict this since the ionized air is more reactive and theoretically will be 1.5x more oxygen-dense at the same time. However, I also anticipate that after testing for combustion in an ionized atmosphere, I will find components of the engine to be oxidized/rusting due to the high reactivity of ozone.

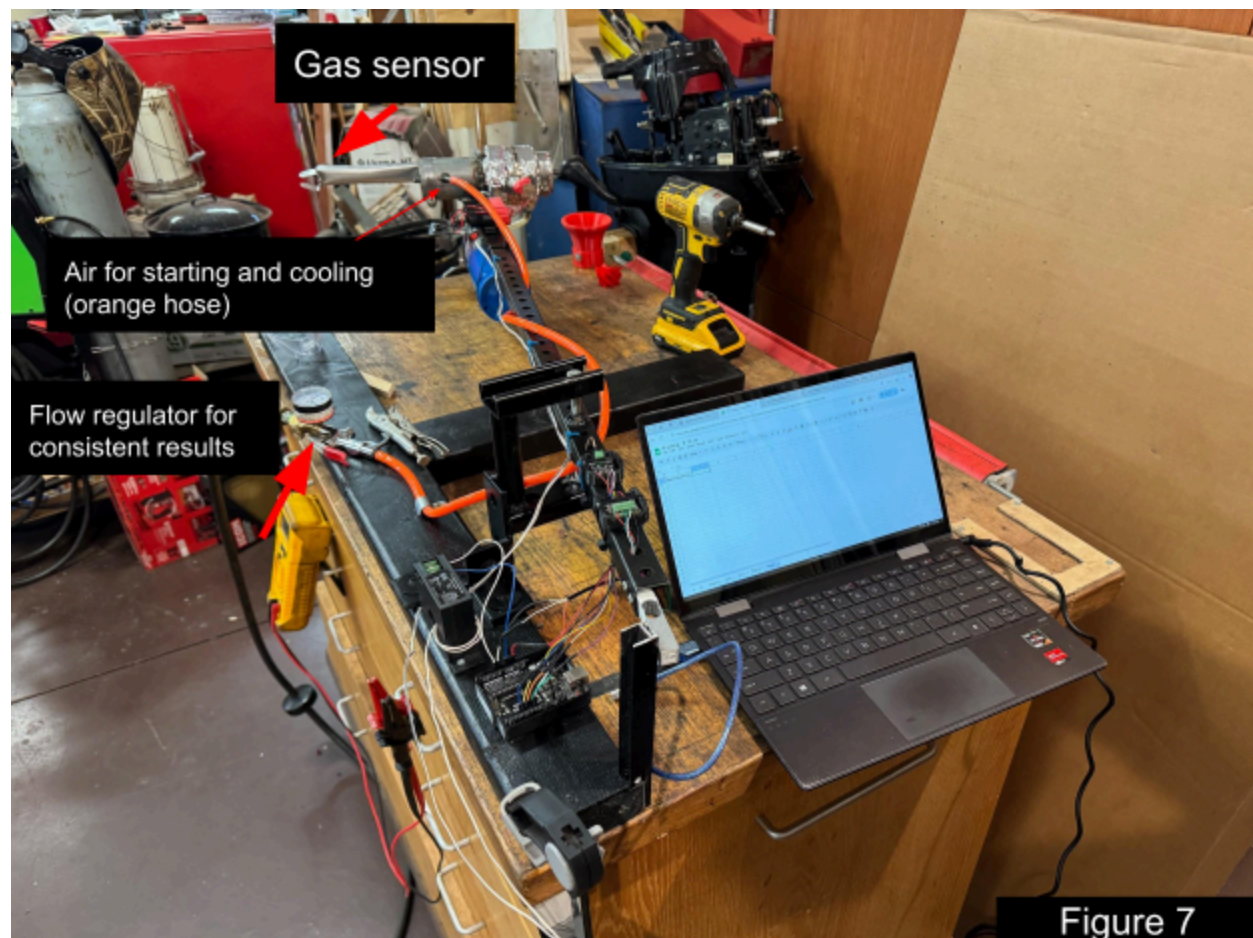
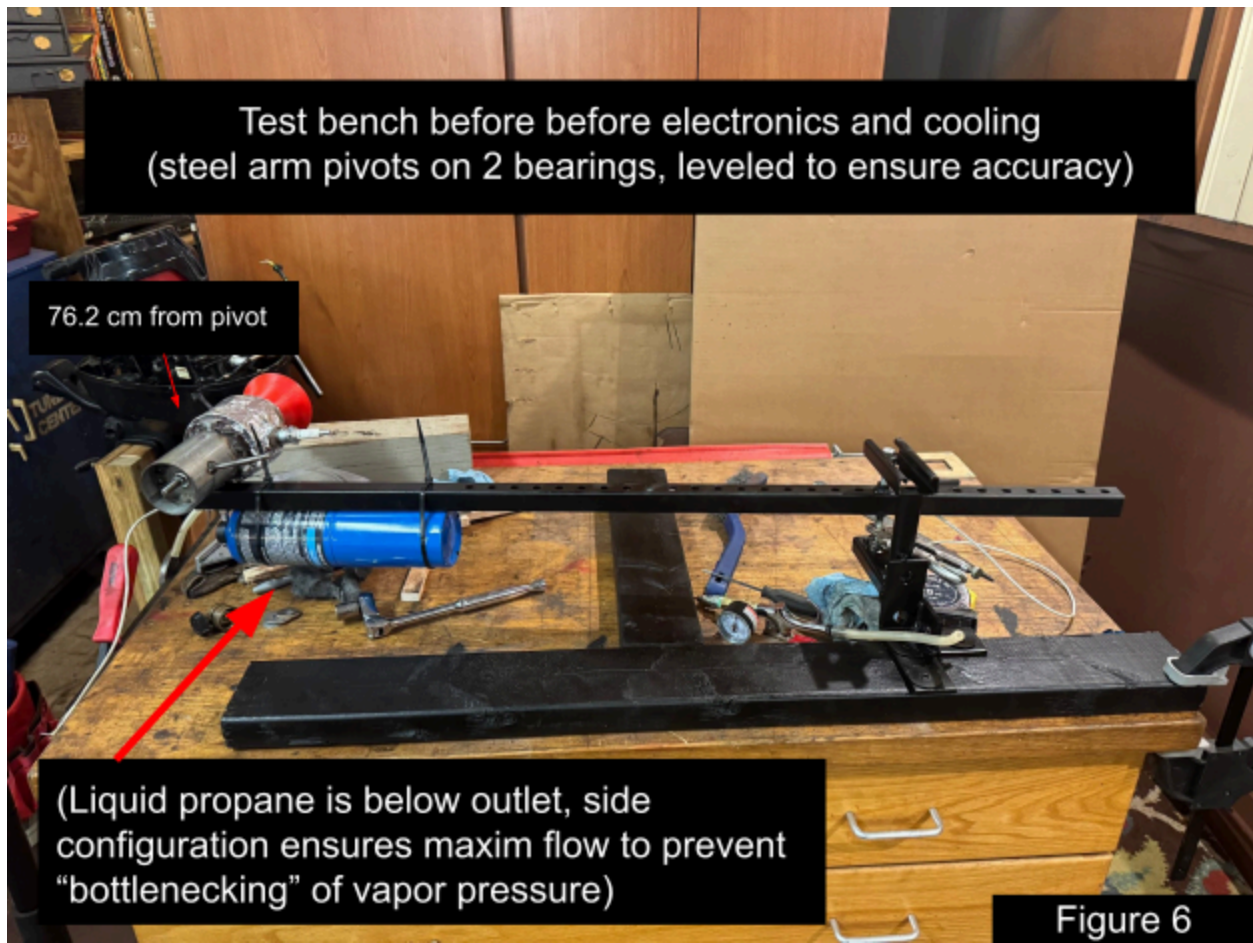
Testing apparatus and other breakdowns are listed below:

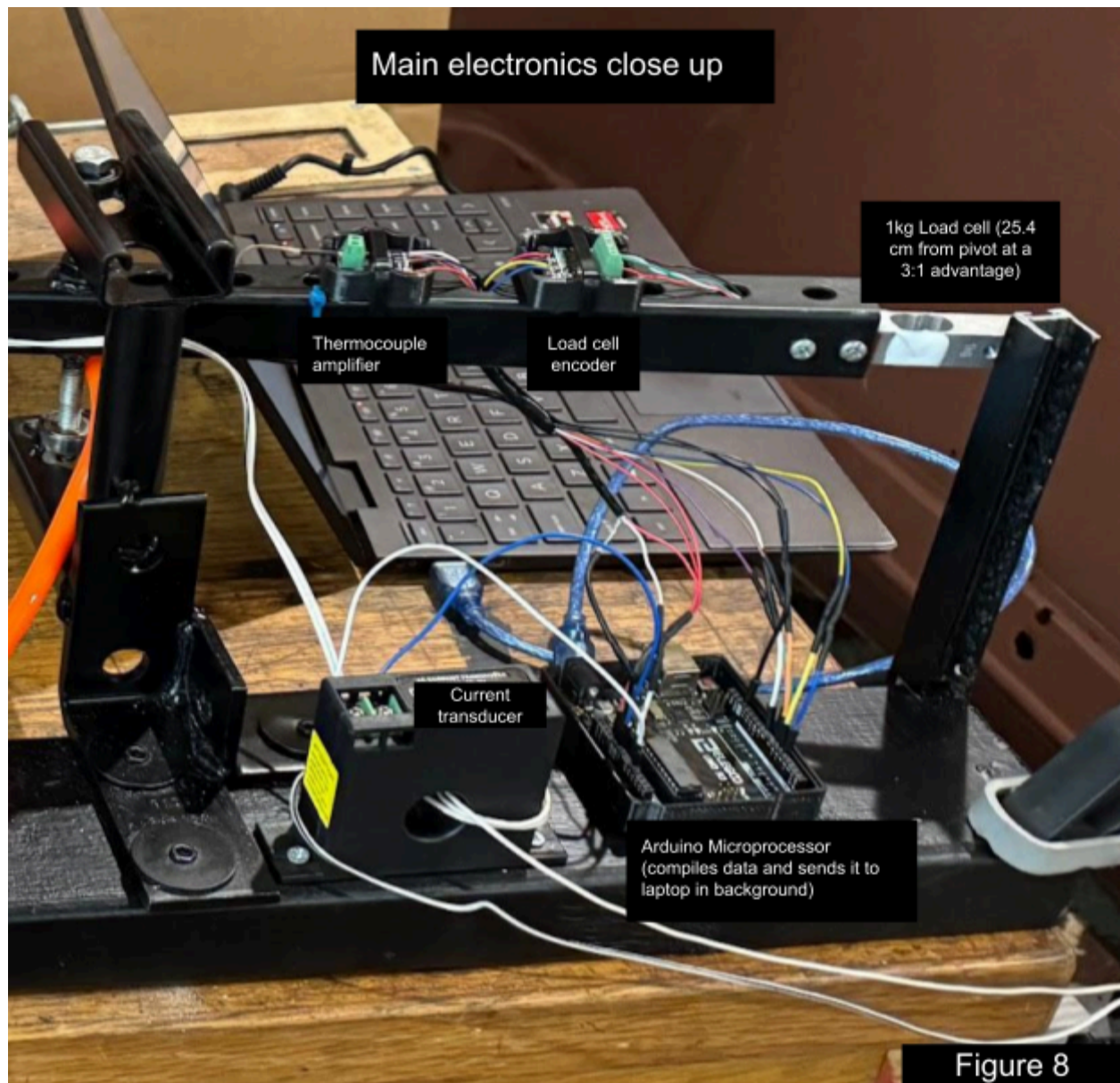


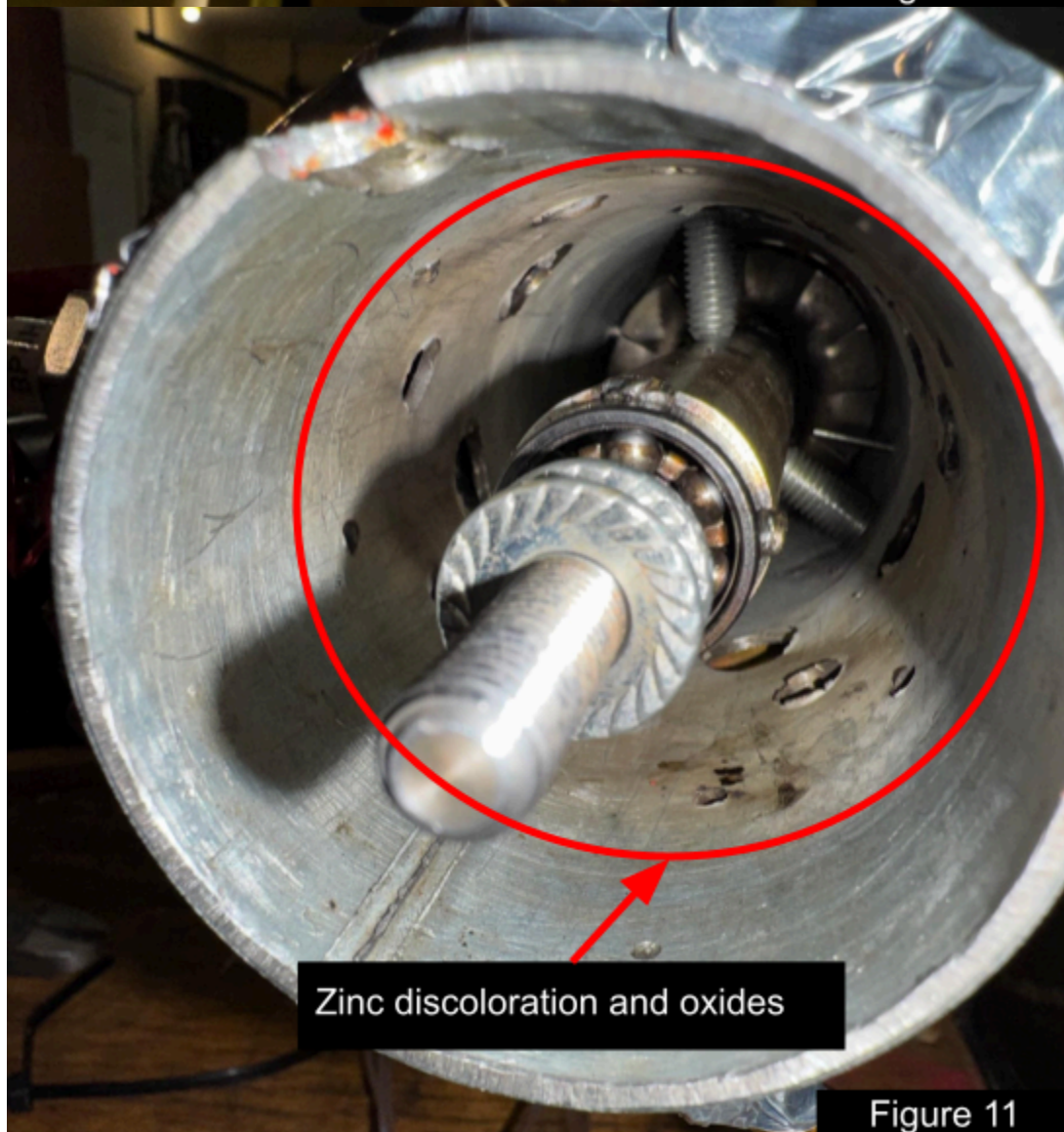
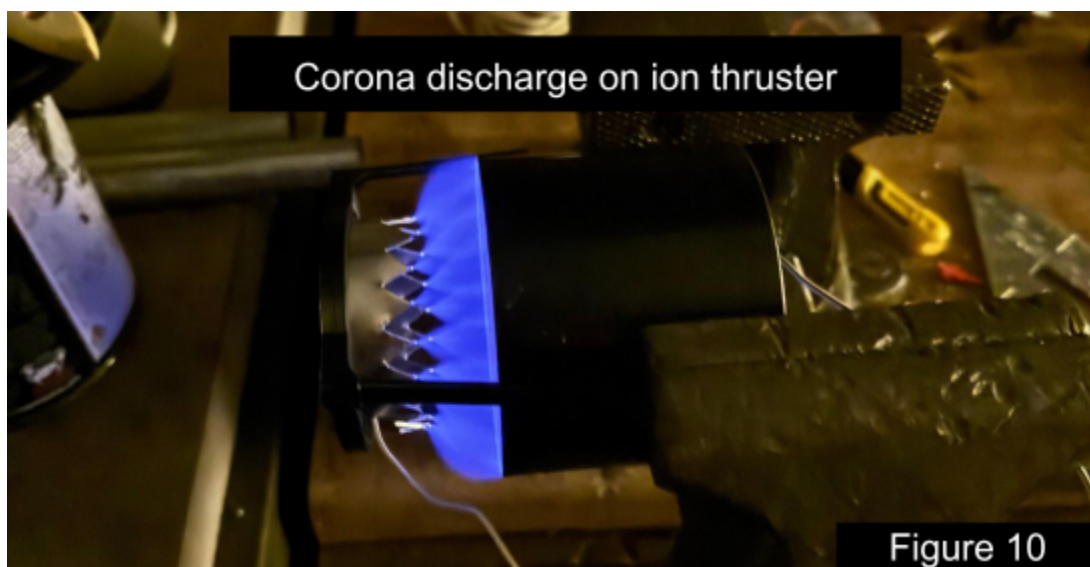
2 Simple mechanical drawing (not drawn for replication) 1





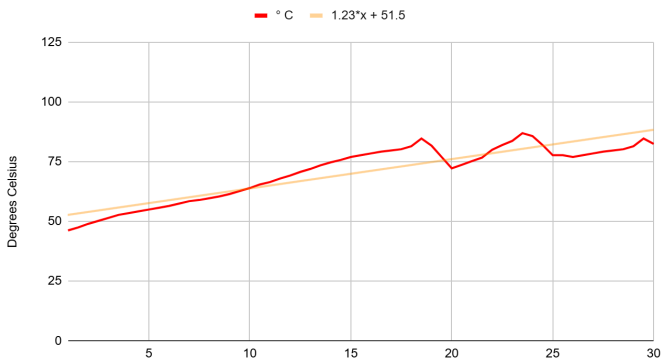




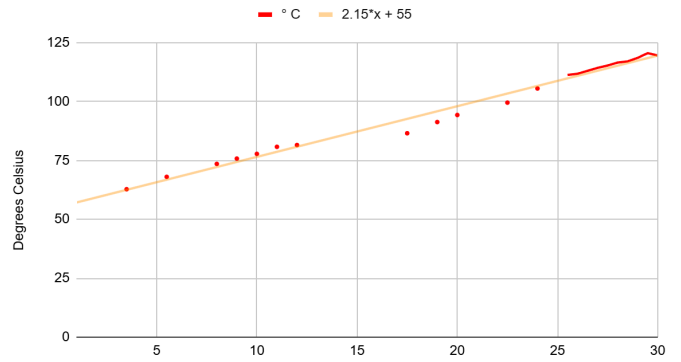


Results: Test 1 left, Test 2 Right

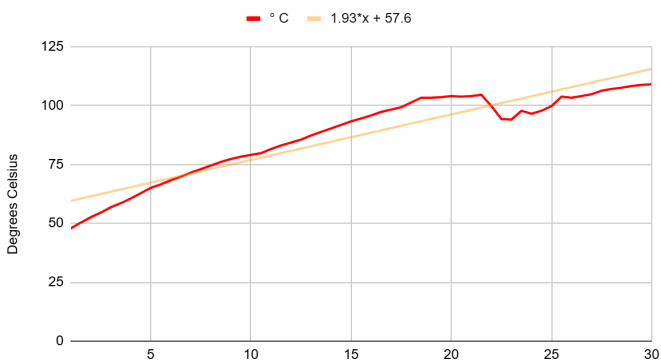
Internal Temp (°Celsius) vs. Time (sec) (Test 1, Trial 1) - Credit: Collin Snider



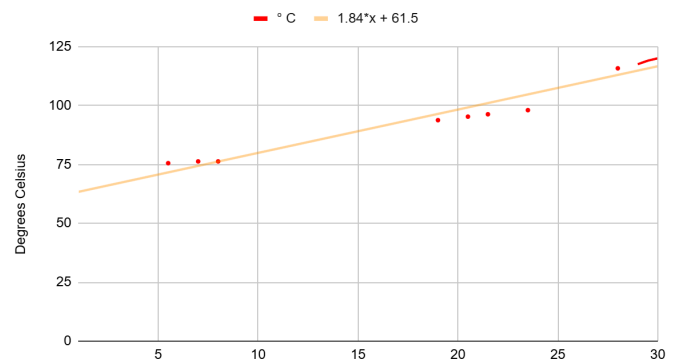
Internal Temp (°Celsius) vs Time (sec) (Test 2, Trial 1) - Credit: Collin Snider



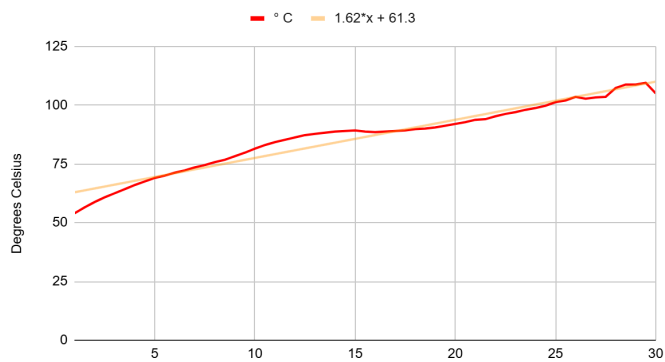
Internal Temp (°Celsius) vs. Time (sec) (Test 1, Trial 2) - Credit: Collin Snider



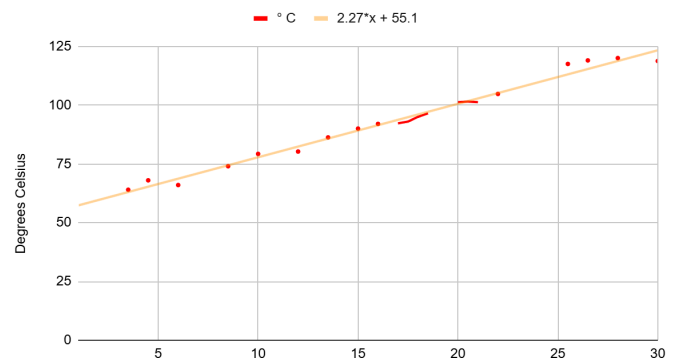
Internal Temp (°Celsius) vs Time (sec) (Test 2, Trial 2) - Credit: Collin Snider



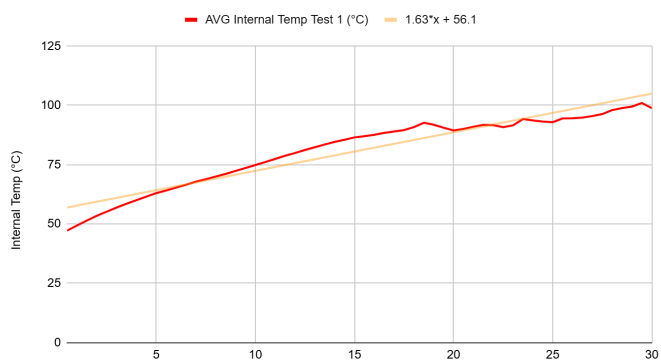
Internal Temp (°Celsius) vs Time (sec) (Test 1, Trial 3) - Credit: Collin Snider



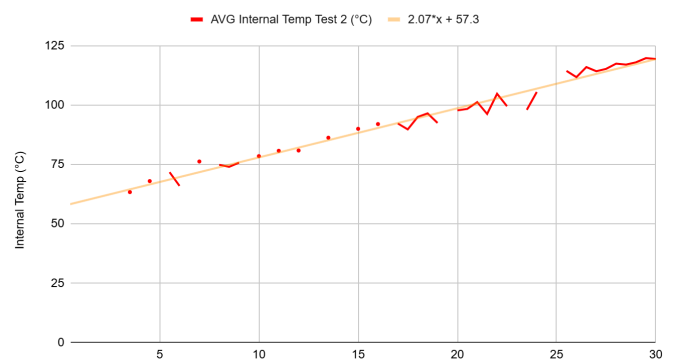
Internal Temp (°Celsius) vs Time (sec) (Test 2, Trial 3) - Credit: Collin Snider



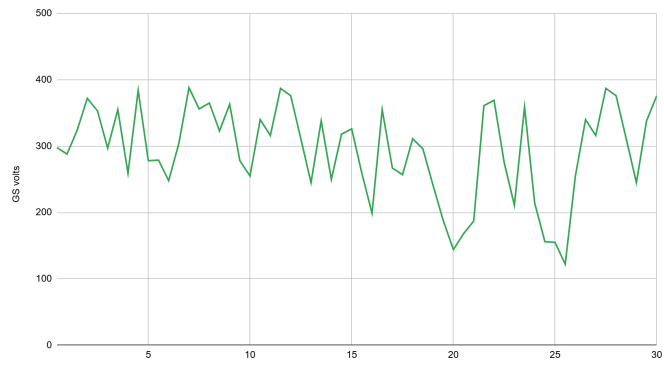
AVG Internal Temp Test 1 (°C) vs. Time (sec) - Credit: Collin Snider



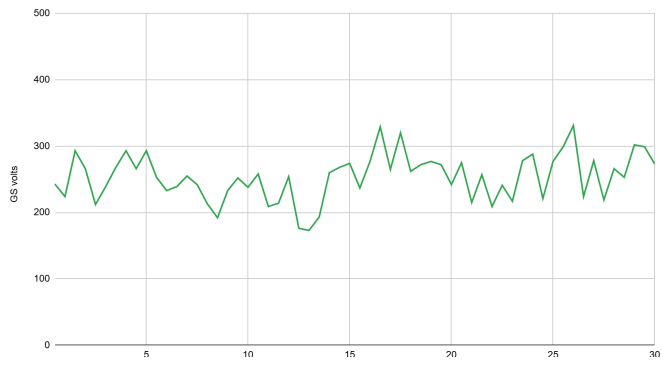
AVG Internal Temp Test 2 (°C) vs. Time (sec) - Credit: Collin Snider



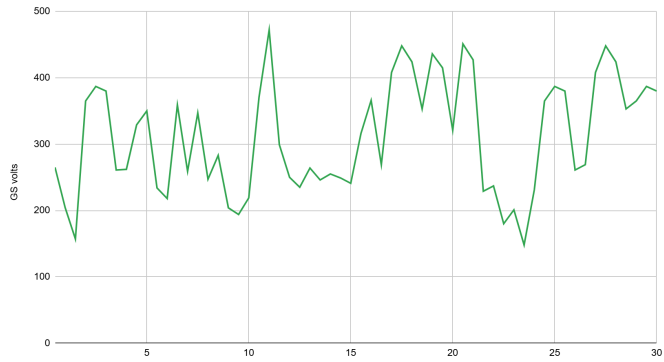
Gas Sensor Reading (0.0001V) vs. Time (sec) (Test 1, Trial 1) - Credit: Collin Snider



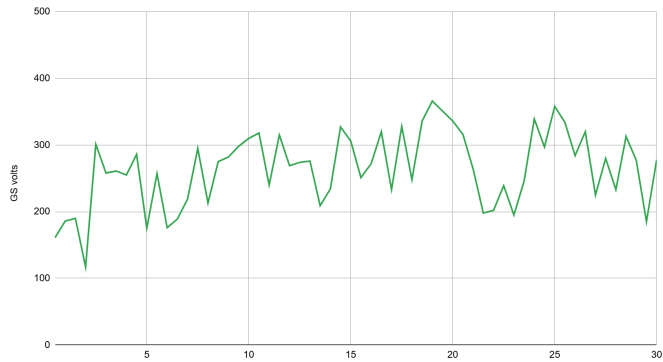
Gas Sensor Reading (0.0001V) vs. Time (sec) (Test 2, Trial 1) - Credit: Collin Snider



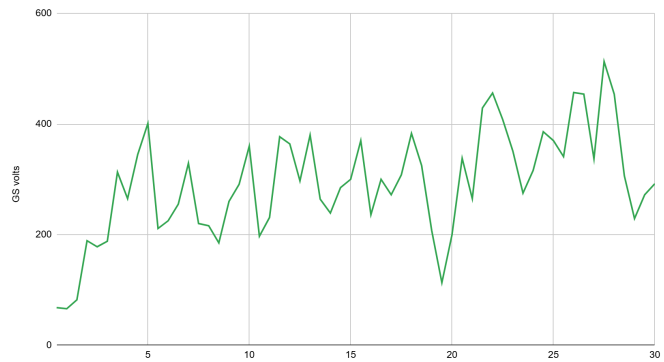
Gas Sensor Reading (0.0001V) vs. Time (sec) (Test 1, Trial 2) - Credit: Collin Snider



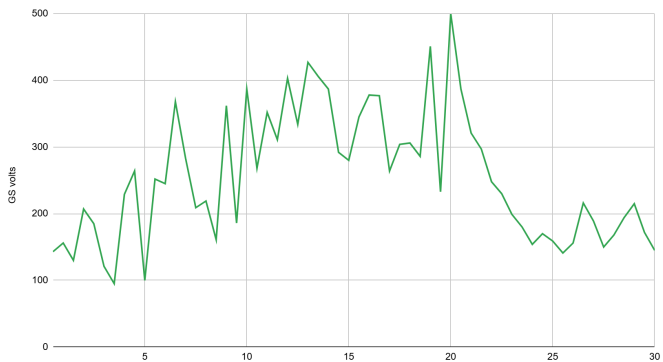
Gas Sensor Reading (0.0001V) vs. Time (sec) (Test 2, Trial 2) - Credit: Collin Snider



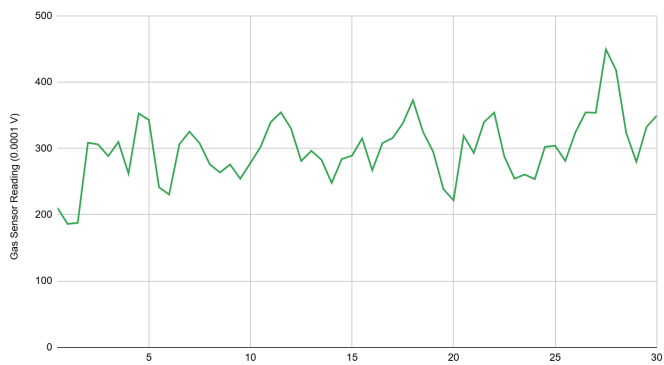
Gas Sensor Reading (0.0001V) vs. Time (sec) (Test 1, Trial 3) - Credit: Collin Snider



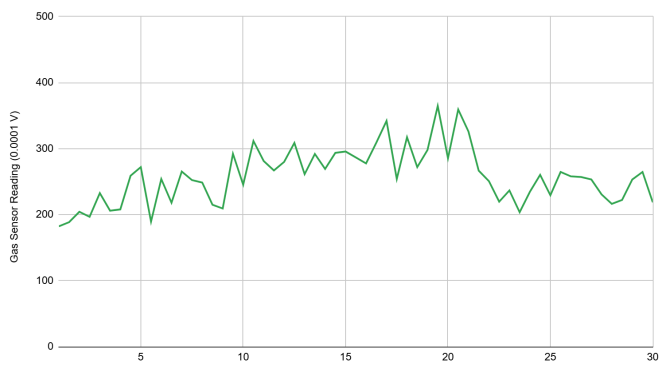
Gas Sensor Reading (0.0001V) vs. Time (sec) (Test 2, Trial 3) - Credit: Collin Snider



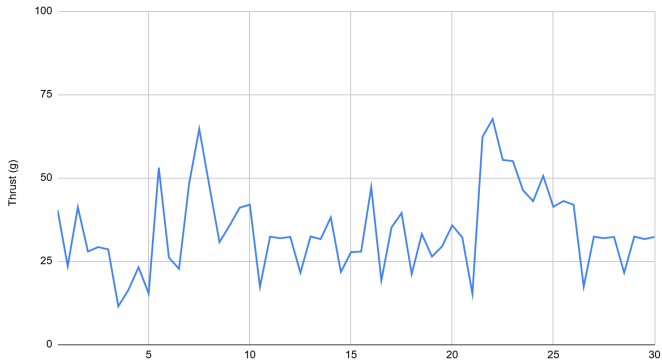
AVG Gas Sensor Reading Test 1 (0.0001 V) vs. Time (sec) - Credit: Collin Snider



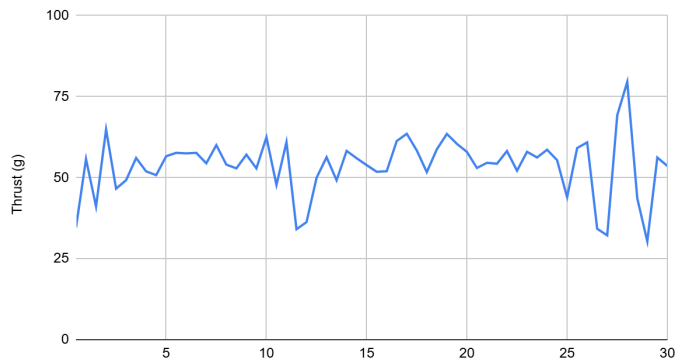
AVG Gas Sensor Reading Test 2 (0.0001 V) vs. Time (sec) - Credit: Collin Snider



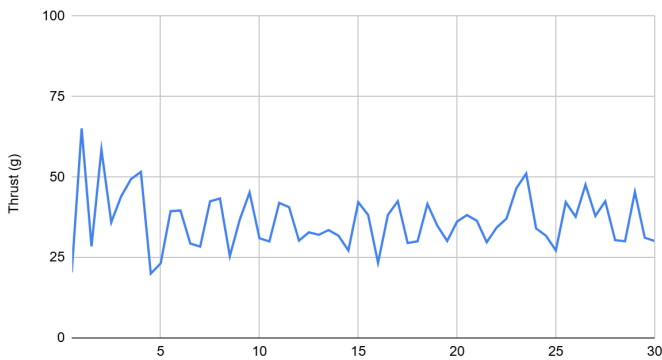
Grams of Thrust vs. Time (sec) (Test 1, Trial 1) - Credit: Collin Snider



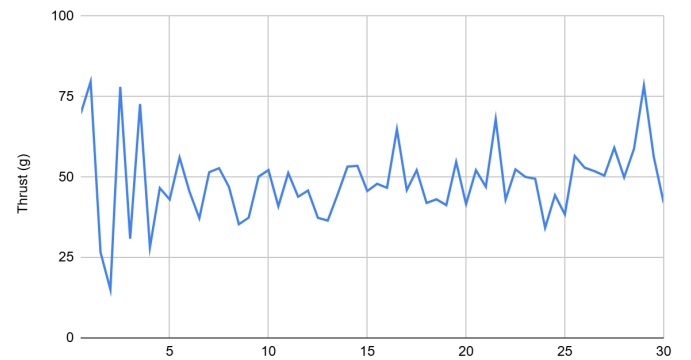
Grams of Thrust vs. Time (sec) (Test 2, Trial 1) - Credit: Collin Snider



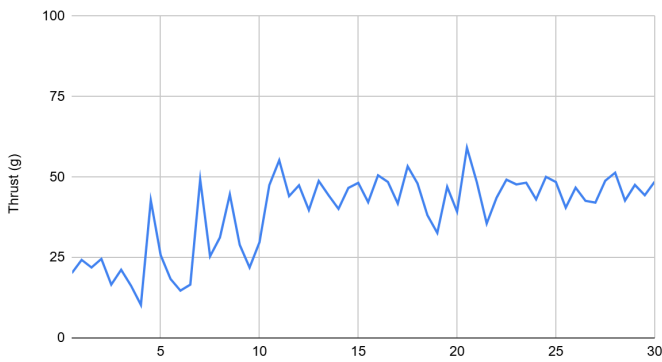
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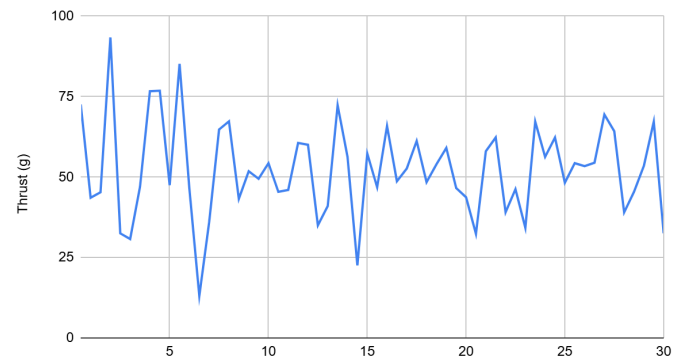
Grams of Thrust vs. Time (sec) (Test 2, Trial 2) - Credit: Collin Snider



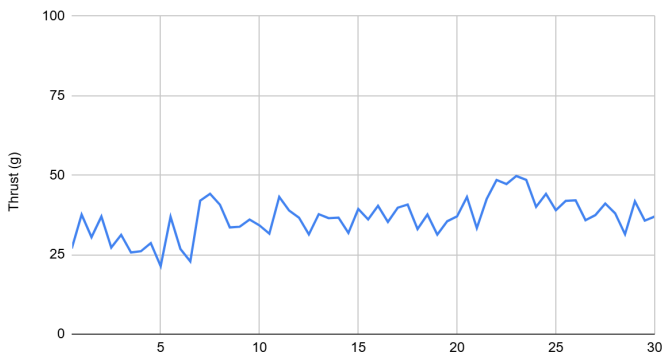
Grams of Thrust vs. Time (sec) (Test 1, Trial 3) - Credit: Collin Snider



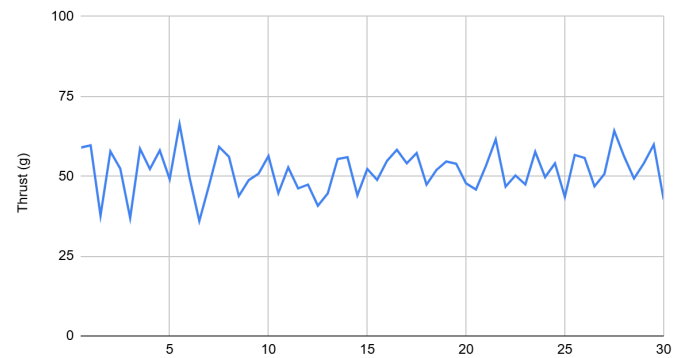
Grams of Thrust vs. Time (sec) (Test 2, Trial 3) - Credit: Collin Snider



AVG Thrust Test 1 (g) vs. Time (sec) - Credit: Collin Snider



AVG Thrust Test 2 (g) vs. Time (sec) - Credit: Collin Snider



After testing, I had to go through and filter the raw data into useful data which can be found in graph form above. Some notable findings from the compilation that I found are increased power produced for Test 2, lower average oxidizing/combustible exhaust value also for Test 2, and a higher rate of internal temperature increase in Test 2. I did not find any of these results overly surprising as my research had led me to form a sound hypothesis. However, the magnitude of power increase was quite astonishing.

To calculate the overall power produced over the three trials of each test I first converted to newton meters, which can then be converted to watt hours or in my case milliwatt hours due to the scale of testing. Test 1 had a total of 35.262 mWh output, and Test 2 showed a total of 49.663 mWh. This is about a 40.84% increase in power; I expected this, perhaps not to this extent. Many patents claim a 7-10% power increase (Varasundharosoth and Phanichamnuay), with parasitic loss of the ionizing components. Although I did measure the draw of the ionizing components, it ended up being 173.317 mWh, which of course is not self-sustaining. With my available equipment and a lack of optimization, this experiment set out to see the drastic effects of such a design (such as exhaust pollution, corrosion, EMI, or temperature changes) which was accomplished. On top of demonstrating that, it also showed how different intake gases may have a major effect on peak power.

A result that I found very pleasing and hard to dispute was the average reading of the gas sensor(MiCS5524) of each test. This value is proprietary and is unable to distinguish the types of gases; however, I was able to differentiate the total change in harmful emissions. One concern going into this project was that the ion components would lead to an increase in NO_x, or ozone emissions since they are produced by the high voltage gradient. However, as predicted, the high temperature of the engine was able to break down each of these into their stable form or a safer byproduct of combustion. The readings output by the sensor for Test 1 averaged a reading of

299.156, and Test 2 averaged a value of 257.744. This shows an improvement of 13.84% in oxidizing gases and VOCs in the exhaust. The reason that I find this value so substantial is that for all trials, the deviation from average was within -2% and +4%, showing great confidence in the consistency.

In contrast some disadvantages of using an ionic thruster in the intake stage of a turbojet engine do exist. The keen-eyed reader may have noticed that temperature readings were inconsistent for Test 2. This is due to the high sensitivity of the thermocouple (0.007 V total variance) and the higher electromagnetic interference (EMI) produced by the 400kV corona discharge a few inches away. Despite my best efforts shielding the wire in a grounded metal braid and the steel tube, many of the data points were recognized as 'Null' by the amplifier. This is likely solvable with better EMI shielding and smarter people developing said shielding. At the conclusion of each trial in Test 2, I made sure to turn off the voltage multipliers to get a few reliable peak temperature readings. This inconsistency is the reason I will be basing the analysis of internal temperature on the slope of the line of best fit and the peak temperatures. With higher combustion efficiency comes increased temperatures, something greatly exemplified in these results. The rate of increase in temp of Test 1 averaged 1.63°C per second, compared to 2.07°C per second in Test 2. This shows a 26.99% increase in the rate of temperature increase, potentially problematic in the real world; however, this is speculation. The higher rate also led to a higher peak temperature in the interval, about 20.58°C higher peak for the tests involving ionized atmosphere. More in-depth analysis of equilibrium temperature and material studies would need to be conducted to further assess what this difference in internal temperature may cause.

Finally, some of my physical observations in testing did not show through the data and the picture of the combustion chamber post-testing (Figure 11). Surprisingly, there was less corrosion than I had expected on carbon steel parts, if fact none visible at all. Further, the discoloration of oxidation of the galvanic zinc coating on the housing can be attributed to high temperatures alone and is not unique to either test.

After the detailed analysis of a combination of over 2,400 data points and further independent research, I have come to a few conclusions. First and most importantly, ionization of the intake air for a combustion engine can certainly have a positive impact on the combustion efficiency and lead to a reduction in harmful emissions. I can assert this due to my personal data collection as well as collections from many inventors filing patents. Second, much like a turbo or supercharger, despite the parasitic loss drawn from the engine, an ionized intake air produced by a high voltage source can still provide a peak power increase due to its higher efficiency, higher reactivity, and volumetric oxygen content. Although some problems may arise from different temperature conditions or the electromagnetic interference produced, I am confident that given proper and in-depth research, teams of engineers may overcome these and even improve the process further.

Works Cited

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